Score-based diffusion: Forward SDE and Reverse ODE

Let $(x_t)_{t\in[0,T]}$ be a diffusion process on $\mathbb{R}^d$ driven by a standard Wiener process (w_t).

Theorem 0.1 (Forward SDE and its marginal evolution): Assume the **forward** diffusion is defined by the Itô SDE

$$\mathrm{d}x_t = f(x_t, t)\,\mathrm{d}t + g(t)\,\mathrm{d}w_t, \quad t \in [0, T],$$

where w_t is a standard Wiener process in $\mathbb{R}^d$. The initial distribution is $x_0 \sim p_0$ (the data distribution), drift $f : \mathbb{R}^d \times [0,T] \to \mathbb{R}^d$, and scalar diffusion coefficient $g : [0,T] \to (0,\infty)$.

Then the marginal density $p_t(x)$ of x_t exists and satisfies the **Fokker–Planck (forward Kolmogorov) PDE**:

$$\partial_t p_t(x) = -\nabla \cdot (f(x,t)p_t(x)) + \frac{1}{2}g(t)^2 \Delta p_t(x), \quad \forall x \in \mathbb{R}^d.$$

Lemma 0.1 (Itô's Lemma for test functions): Let x_t be the process defined above and $\varphi : \mathbb{R}^d \to \mathbb{R}$ be a smooth function (C^2). The stochastic differential of $\varphi(x_t)$ is given by:

$$\mathrm{d}\varphi(x_t) = \left[\nabla\varphi(x_t)\cdot f(x_t,t) + \frac{1}{2}g(t)^2\Delta\varphi(x_t)\right]\mathrm{d}t + g(t)\nabla\varphi(x_t)\cdot \mathrm{d}w_t.$$

Proof: Let $\varphi(x)$ be an arbitrary smooth test function with compact support ($\varphi \in C_c^\infty(\mathbb{R}^d)$).

We define the expectation $E(t) := \mathbb{E}[\varphi(x_t)]$. Integrating the SDE form given in Lemma 0.1 from 0 to t and taking the expectation:

$$\begin{aligned}\mathbb{E}[\varphi(x_t)] - \mathbb{E}[\varphi(x_0)] = \mathbb{E}&\left[\int_0^t \left[f(x_s,s)\cdot\nabla\varphi(x_s) + \frac{1}{2}g(s)^2\Delta\varphi(x_s)\right]\mathrm{d}s\right] \\ &+\mathbb{E}\left[\int_0^t g(s)\nabla\varphi(x_s)\cdot \mathrm{d}w_s\right].\end{aligned}$$

The second term (the stochastic integral) vanishes because the Itô integral with respect to a Wiener process is a martingale (and has zero expectation starting from 0).

Differentiating the remaining term with respect to t:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[\varphi(x_t)] = \mathbb{E}\left[f(x_t,t)\cdot\nabla\varphi(x_t) + \frac{1}{2}g(t)^2\Delta\varphi(x_t)\right].$$

Now, we rewrite the expectation using the probability density $p_{t(x)}$:

$$\int_{\mathbb{R}^d}\varphi(x)\partial_t p_{t(x)}\,\mathrm{d}x = \int_{\mathbb{R}^d}\left[f(x,t)\cdot\nabla\varphi(x) + \frac{1}{2}g(t)^2\Delta\varphi(x)\right]p_{t(x)}\,\mathrm{d}x.$$

We apply integration by parts to move the derivatives from the test function φ to the density p_t. Since φ has compact support, the boundary terms vanish.

1. For the drift term:

$$\int (f\cdot\nabla\varphi)p_t\,\mathrm{d}x = -\int \varphi\nabla\cdot(fp_t)\,\mathrm{d}x.$$

2. For the diffusion term (applying IBP twice):

$$\int (\Delta\varphi)p_t\,\mathrm{d}x = \int \varphi\Delta p_t\,\mathrm{d}x.$$

Substituting these back yields:

$$\int_{\mathbb{R}^d} \varphi(x)\partial_t p_{t(x)}\, \mathrm{d}x = \int_{\mathbb{R}^d} \varphi(x)\left[-\nabla \cdot \left(f(x,t)p_{t(x)}\right) + \frac{1}{2}g(t)^2 \Delta p_{t(x)}\right] \mathrm{d}x.$$

Rearranging terms to one side:

$$\int_{\mathbb{R}^d} \varphi(x)\left[\partial_t p_{t(x)} + \nabla \cdot (f p_t) - \frac{1}{2}g(t)^2 \Delta p_t\right] \mathrm{d}x = 0.$$

Since this equality holds for **any** smooth test function φ with compact support, the fundamental lemma of calculus of variations implies that the term in the brackets must be zero almost everywhere. Thus, p_t satisfies the Fokker–Planck equation. ■

Definition 0.1 (Score of the marginal): Define the **score** function as the gradient of the log-density:

$$s_t(x) := \nabla_x \log p_t(x) = \frac{\nabla p_{t(x)}}{p_{t(x)}}.$$

Theorem 0.2 (Reverse-time SDE): Assume p_t is smooth and strictly positive for $t \in (0, T]$. Let $\overline{w}_t$ be a standard Wiener process under time reversal.

Then the **reverse-time** process that runs from $t = T$ down to $t = 0$ is governed by the SDE:

$$\mathrm{d}x_t = \left[f(x_t, t) - g(t)^2 s_t(x_t)\right] \mathrm{d}t + g(t)\, \mathrm{d}\overline{w}_t,$$

integrated backward in time from $T \to 0$.

Proof: Consider the general form of the reverse drift $\tilde{f}(x,t)$. For a diffusion process with scalar diffusion $g(t)$, the relationship between the forward drift f and the reverse drift $\tilde{f}$ is given by:

$$\tilde{f}(x,t) = f(x,t) - \frac{1}{p_{t(x)}}\nabla \cdot \left(D(t)p_{t(x)}\right),$$

where $D(t) = g(t)^2 I$ is the diffusion tensor. Substituting $D(t)$ into the equation:

$$\begin{aligned}\tilde{f}(x,t) &= f(x,t) - \frac{1}{p_{t(x)}}\nabla \cdot \left(g(t)^2 p_{t(x)}\right) \\ &= f(x,t) - g(t)^2 \frac{\nabla p_{t(x)}}{p_{t(x)}}.\end{aligned}$$

Using the definition of the score $s_{t(x)} = \nabla \log p_{t(x)} = \frac{\nabla p_{t(x)}}{p_{t(x)}}$, we obtain:

$$\tilde{f}(x,t) = f(x,t) - g(t)^2 s_{t(x)}.$$

Thus, the reverse SDE has the modified drift $f - g^2 s_t$ and the same diffusion coefficient $g(t)$. ■

Theorem 0.3 (Probability flow ODE): Under the same assumptions, consider the deterministic ODE:

$$\frac{\mathrm{d}x_t}{\mathrm{d}t} = f(x_t, t) - \frac{1}{2}g(t)^2 s_t(x_t),$$

integrated backward in time from $T \to 0$.

Then this **probability flow ODE** has the **same marginal densities** as the forward SDE.

Proof: We show that the probability density evolved by this ODE satisfies the same Fokker–Planck equation as the SDE. Recall the Fokker–Planck equation for the SDE from Theorem 1:

$$\partial_t p_t = -\nabla \cdot (f p_t) + \frac{1}{2} g(t)^2 \Delta p_t.$$

We use the identity $\Delta p_t = \nabla \cdot (\nabla p_t)$. By definition of the score, $\nabla p_t = p_t s_t$. Therefore, we can rewrite the diffusion term as:

$$\frac{1}{2} g(t)^2 \Delta p_t = \frac{1}{2} g(t)^2 \nabla \cdot (p_t s_t) = \nabla \cdot \left(\frac{1}{2} g(t)^2 s_t p_t \right).$$

Substituting this back into the Fokker–Planck equation:

$$\begin{aligned} \partial_t p_t &= -\nabla \cdot (f p_t) + \nabla \cdot \left(\frac{1}{2} g(t)^2 s_t p_t \right) \\ &= -\nabla \cdot \left[\left(f - \frac{1}{2} g(t)^2 s_t \right) p_t \right]. \end{aligned}$$

This equation is exactly the **continuity equation** (Liouville equation) describing the evolution of densities under a deterministic flow field $v_{t(x)}$:

$$\partial_t p_t + \nabla \cdot (v_t p_t) = 0.$$

Comparing terms, we identify the velocity field:

$$v_{t(x)} = f(x, t) - \frac{1}{2} g(t)^2 s_{t(x)}.$$

Since the ODE $\frac{\mathrm{d}x_t}{\mathrm{d}t} = v_{t(x_t)}$ implies the continuity equation for its marginals, and this equation is identical to the SDE's Fokker–Planck equation, the ODE and the SDE share the same marginal densities p_t. ■